

Integrating spatial transcriptomics data across different conditions, technologies and developmental stages

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With the rapid generation of spatial transcriptomics (ST) data, integrative analysis of multiple ST datasets from different conditions, technologies and developmental stages is becoming increasingly important. Here we present a graph attention neural network called STAligner for integrating and aligning ST datasets, enabling spatially aware data integration, simultaneous spatial domain identification and downstream comparative analysis. We apply STAligner to ST datasets of the human cortex slices from different samples, the mouse olfactory bulb slices generated by two profiling technologies, the mouse hippocampus tissue slices under normal and Alzheimer's disease conditions, and the spatiotemporal atlases of mouse organogenesis. STAligner efficiently captures the shared tissue structures across different slices, the disease-related substructures and the dynamical changes during mouse embryonic development. In addition, the shared spatial domain and nearest-neighbor pairs identified by STAligner can be further considered as corresponding pairs to guide the three-dimensional reconstruction of consecutive slices, achieving more accurate local structure-guided registration than the existing method.

In multicellular organisms, cells are deeply influenced by their tissue environment. Knowledge of the spatial position of cells has a critical role in understanding why cells from different locations have distinctive structures and functions¹. Recent spatial transcriptomics (ST) technologies provide new ways to measure messenger RNA expression and corresponding spatial coordinates in tissues². Various ST technologies have been developed with different spatial resolutions. For example, the spatial barcoding-based method 10x Visium achieves a spatial resolution of 55 μm with each capture site (that is, spot) containing approximately 1–10 cells³. Slide-seq improves spatial resolution to near-cellular level (10 μm)^{4,5} and Stereo-seq further reaches subcellular resolution (0.22 μm)⁶. Indeed, these technologies have been widely used to explore the biological architecture of tissues^{4,5,7}, the biomarkers of complex disease⁸ and the development process of organogenesis⁶.

In particular, computational methods have been successfully applied to ST data of a single condition, technology or developmental stage to characterize the tissue structure^{9–12}, identify spatially variable genes^{13,14} and decipher spatial cell–cell communication patterns¹⁵. However, integrated analysis of ST datasets produced by different conditions, technologies or developmental stages remains challenging.

Identifying the shared and specific spatial domains (defined as regions with similar spatial expression patterns) is one of the fundamental tasks for the integrative analysis of multiple ST datasets. Several methods have been proposed to address this issue in single-slice ST datasets by incorporating spatial information and gene expression profiles^{10–12}. For example, STAGATE¹² uses a deep learning model and leverages the graph attention mechanism to learn the spatial heterogeneity. However, integrative analysis of multiple ST datasets poses a

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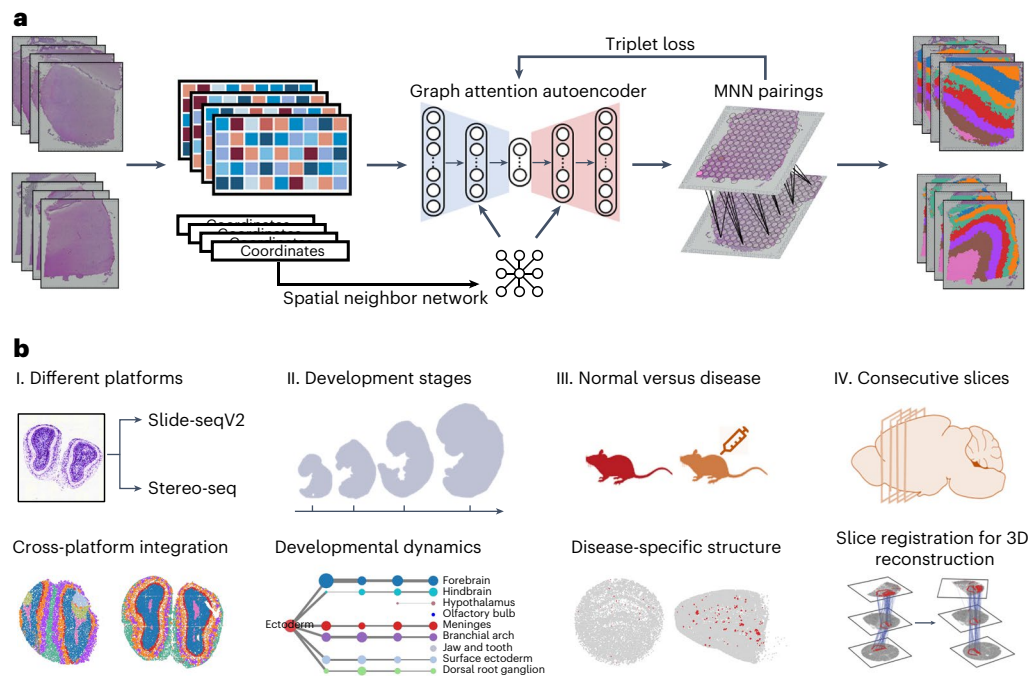


Fig. 1 | Overview of STAligner. a, STAligner first normalizes the expression profiles for all spots and constructs a spatial neighbor network using the spatial coordinates. STAligner further employs a graph attention autoencoder neural network to extract spatially aware embedding, and constructs the spot triplets based on current embeddings to guide the alignment process by attracting similar spots and discriminating dissimilar spots across slices. STAligner introduces the triplet loss to update the spot embedding to reduce

the distance from the anchor to positive spot, and increase the distance from the anchor to negative spot. The triplet construction and autoencoder training are optimized iteratively until batch-corrected embeddings are generated. **b**, STAligner can be applied to integrate ST datasets to achieve alignment and simultaneous identification of spatial domains from different biological samples in **a**, technological platforms (I), developmental (embryonic) stages (II), disease conditions (III) and consecutive slices of a tissue for 3D slice alignment (IV).

unique challenge, as the batch effects across different datasets may mask the actual biological signals and hamper data integration.

Compared with single-slice ST data analytics for deciphering spatial heterogeneity, integrative analysis of multiple ST datasets can provide more comprehensive characterizations of spatial tissue structures. For example, integrating slices from normal and diseased tissue can facilitate the identification of disease-specific spatial domains¹⁶. However, most current integration analyzes either pool gene expression data across slices or employs existing batch correction methods developed for single-cell RNA sequencing (scRNA-seq) data, such as Harmony¹⁷ and Seurat V3¹⁸, without considering the spatial coordinates. An unsupervised spatial embedding method, SEDR⁹, was combined with Harmony to embed ST datasets for batch correction. As these two modules are optimized independently, their combined performance is limited. A recently developed method, PASTE¹⁹, integrates adjacent slices into a center slice to improve the clustering performance of spatial domains and achieve three-dimensional (3D) reconstruction. However, PASTE requires that adjacent slices are biological replicates with global and rigid similarity, which is often violated in heterogeneous tissue.

To this end, we develop a graph attention neural network, named STAligner, for integrating multiple ST datasets through generating batch-corrected spot embeddings, enabling consensus spatial domain identification, downstream comparative analysis and 3D tissue reconstruction. We demonstrate improved performance of STAligner on diverse datasets from different individuals, technologies, disease conditions and developmental stages.

Results

Overview of STAligner

STAligner takes the normalized gene expression and spatial coordinates of ST data from multiple tissue slices as input (Fig. 1). For each ST slice,

STAligner constructs a spatial neighbor graph between spots using the spatial coordinates. STAligner employs a graph attention autoencoder neural network to learn spatially aware embeddings based on the spatial neighbor graph, and constructs the spot triplets based on current embeddings to guide the integration and alignment process of different slices by attracting similar spots and discriminating dissimilar spots across slices. STAligner defines the triplet that contains anchor-positive and anchor-negative spot pairs. The former is defined as the mutual nearest neighbors (MNNs) with similar gene expressions but belong to two different slices, and the latter belongs to the same slice with different spatial positions and dissimilar expressions. STAligner introduces the triplet loss to update the spot embedding to reduce the distance from the anchor spot to the positive one, and increase the distance from the anchor spot to the negative one. The graph attention autoencoder training and triplet construction are optimized iteratively until batch-corrected embeddings are generated.

Improved integration on slices from the same platform

To quantitatively evaluate the integration performance of STAligner, we applied it to the human dorsolateral prefrontal cortex (DLPFC) ST dataset⁷ generated by 10x Visium. It consists of 12 tissue slices from 3 adult samples with 4 adjacent slices for each. The original study has manually annotated 6 neocortical layers from layer 1 to layer 6 and the white matter (WM) (Fig. 2a). By taking the manual annotations as ground truth, we evaluated the performance of STAligner on simultaneous identification of spatial domains and compared it with that of the non-spatial integration method Harmony and two recently developed spatial integration approaches SEDR and PASTE in terms of the adjusted Rand index (ARI) (Methods).

We first evaluated the performance on adjacent slices of a single sample and observed that both STAligner and SEDR can effectively integrate the four slices with matched low-dimensional embeddings

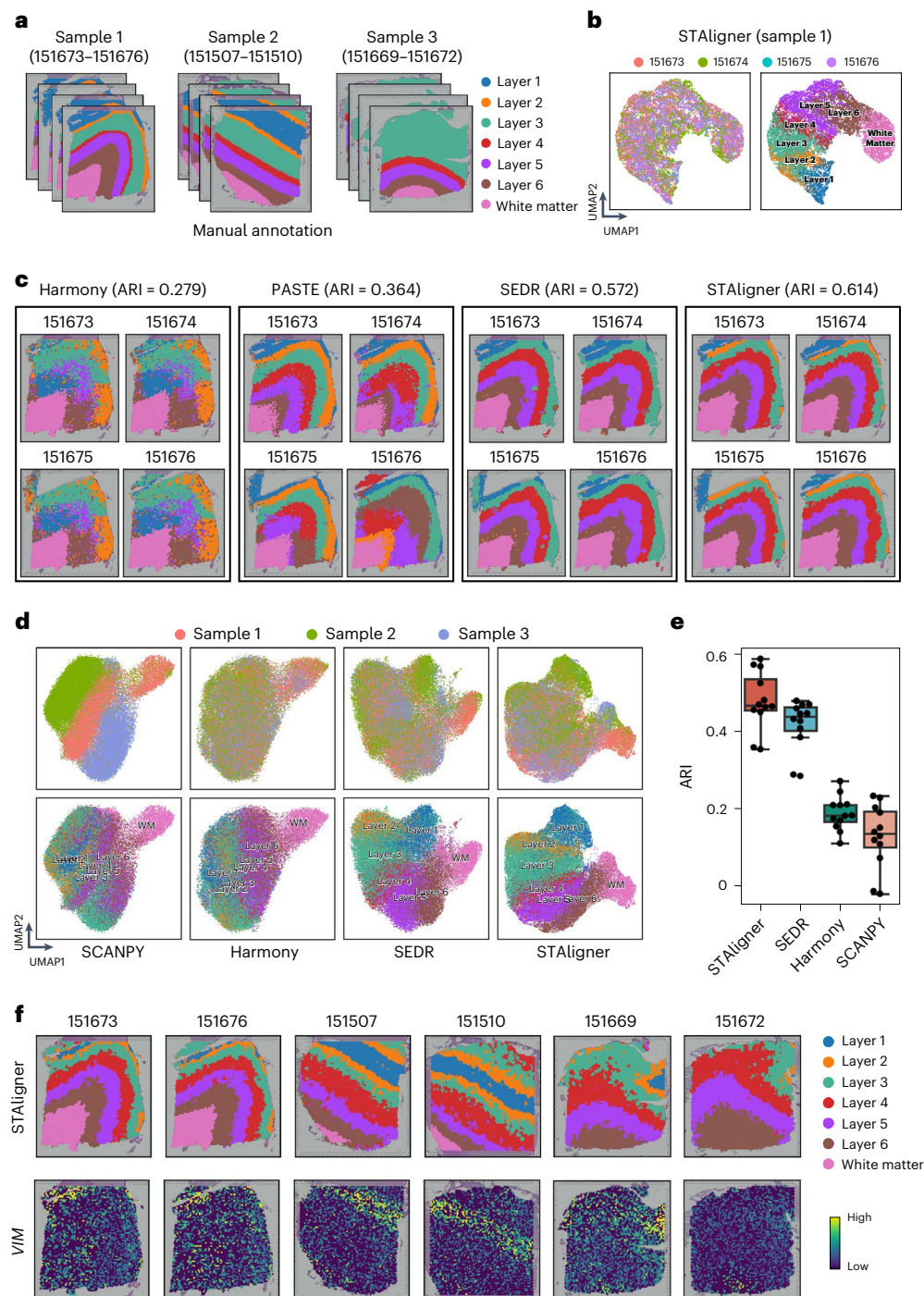


Fig. 2 | Benchmarking on the human DLPFC slices from the same and different samples. **a**, Manual annotations of cortical layers and white matter on the slices of three DLPFC samples. **b**, UMAP plots of STAligner embeddings colored by slices (left) and cortical layers (right) for integrating and aligning four slices of sample 1. **c**, Aligned spatial domain identified simultaneously by integrating four slices of sample 1 via Harmony, PASTE, SEDR, and STAligner respectively. **d**, UMAP plots of embeddings colored by slices (top) and cortical layers (bottom) by integrating and aligning 12 slices of all 3 samples via SCANPY, Harmony, SEDR

and STAligner respectively. **e**, Boxplot overlaid with swarm plot for the clustering accuracy of spatial domains identified simultaneously by the four methods for all 12 slices in terms of ARI scores. In the boxplot, the center line, box limits and whiskers denote the median, upper and lower quartiles, and 1.5× interquartile range, respectively. The swarm plot shows the distribution of accuracy of all slices. **f**, Spatial domains identified simultaneously by STAligner (top) and spatial expression of marker gene *VIM* for layer 1 (bottom) in six slices respectively.

(Fig. 2b and Supplementary Fig. 1a). The common cortical layers across slices were evenly mixed, and different layers were separated and ordered with spatial structure from layer 1 to layer 6 and white matter. Although Harmony cannot separate the cortical layers, its embeddings of the four slices were mixed. PASTE captures the layer structures but the slices were not evenly mixed as it did not perform embedding

alignment explicitly. STAligner achieved the highest ARI scores among all methods for the three samples. For example, STAligner accurately captured the layer boundaries and obtained a higher clustering accuracy (ARI = 0.61) than that of Harmony (ARI = 0.28), PASTE (ARI = 0.36) and SEDR (ARI = 0.57) for sample 1 (slices 151673–151676, Fig. 2c). We find that the layer structure identified by the non-spatial approach

Harmony suffered from serious noisy boundary issues and could only identify the layer 3 and white matter structure correctly. For comparison, the spatially aware embedding methods could more accurately capture tissue architecture heterogeneity and the obtained domain boundary was notably smoother. Similar conclusions can also be found for the other two samples (151507–151510 and 151669–151672), indicating the importance of considering spatial information for integration of multiple ST slices even from the same tissue sample (Supplementary Fig. 1b). We also found that STAligner can identify consensus domains across slices missed by non-integration methods. For example, the laminar structure of layer 4 in slices 151676 and 151507 was missed by STAGATE and BayesSpace (Supplementary Fig. 2). In addition, we further tested the robustness of STAligner regarding read-depth variation by downsampling experiments on DLPFC data to simulate different read depths, and found that STAligner showed robust clustering performance even when the gene expressions were downsampled at a very sparse level (downsampling rate < 0.5) (Supplementary Fig. 3).

We further compared the performance on a more challenging setting by integrating all 12 slices from the 3 samples. The Uniform Manifold Approximation and Projection (UMAP) plot of unaligned embeddings produced by a tool developed for Single-Cell Analysis in Python (SCANPY) showed that the three samples exhibited substantial batch effects with common cortical layers from different samples separated (Fig. 2d). The embeddings of Harmony and SEDR were overcorrected with a disordered layer structure, whereas STAligner evenly mixed the spots from the shared layers across samples and made the layers organized as their spatial locations. This conclusion can be further confirmed by quantitative measure of batch-corrected embedding on both real and simulation datasets (Supplementary Figs. 4a and 5), where STAligner achieved a comparable divergence score but a much higher silhouette coefficient than competing methods. Moreover, the previously reported layer-marker genes⁷, such as *FABP7* (layer 1), *HPCAL1* (layer 2), *CARTPT* (layer 3), *PCP4* (layer 5), *KRT17* (layer 6) and *MBP* (white matter), showed differential expressions among the identified layers (Supplementary Fig. 4b). STAligner demonstrated more consistent spatial domains across samples (Supplementary Fig. 4c) and a higher median ARI score of 0.47, compared with Harmony (median ARI = 0.19) and SEDR (median ARI = 0.43) (Fig. 2e). In addition, STAligner recognized the structure of layer 1 and layer 2 at the top right part of layer 3 in sample 3, which were missed when analyzing this sample alone and can be verified by the expression of marker gene *VIM* (Fig. 2f), demonstrating that STAligner can depict the shared tissue structures across different samples.

Some ST technologies may be limited by the size of the captured area and cannot cover a whole organ of interest for some cases. As a solution, the tissue sample is dissected into multiple slices. We applied STAligner to a mouse brain dataset where the tissue was divided into the sagittal anterior and posterior slices (Supplementary Fig. 6a). STAligner captured the known brain tissue structures of the olfactory bulb (domains 3, 17, 37), cortex (domains 6, 9, 10, 16, 20, 33, 35), stratum (domains 2, 12, 34), thalamus (domains 4, 24, 27, 30), hypothalamus (domains 5, 7, 28), hippocampus (domains 14, 19, 31, 36), cerebellum (domains 8, 11, 22, 32), midbrain (domains 18, 23), hindbrain (domains 1, 13, 25, 29, 39) and corpus callosum (domains 14, 21) (Supplementary Fig. 6b). In particular, STAligner accurately recognized three small structures in the hippocampus region, with domain 31 denoting the dentate gyrus structure, domain 14 denoting cornu ammonis 1 (CA1) and domain 36 denoting CA3 (Supplementary Fig. 6c). We also found that the tissue structures flanking the shared border of the anterior and posterior slice were accurately identified as the same domains, such as the structures CA1 and CA3, and the superficial and deep cortex layer (domains 10, 16, 20), which were also characterized by some marker genes such as *Fibcd1* in CA1, *Cabp7* in CA3²⁰, *Lamp5* in the superficial cortex layer⁷ and *Fzf2* in the deep cortex layer²¹. Thus, STAligner could stitch multiple tissue slices into a large one for deciphering its global structure.

Improved integration on slices from different platforms

Here we applied STAligner to integrate two mouse olfactory bulb ST slices profiled by the Slide-seqV2⁵ and Stereo-seq^{6,9} platforms (Fig. 3a) with clear tissue structures (Fig. 3b) but substantial batch effects (Supplementary Fig. 7). Harmony followed with Louvain clustering failed to recover the laminar organization and SEDR found some intermixed spatial clusters with unclear tissue structures (Fig. 3c). By contrast, STAligner can decipher the known tissue structures according to the annotated laminar structure in the DAPI-stained image⁹ and Allen Mouse Brain Atlas annotation (Fig. 3b) from both slices, indicating these two platforms indeed capture most of the key tissue structures. The six shared spatial domains by the two slices were spatially ordered from the inner layers, middle layers to outer layers of the mouse olfactory bulb, including the rostral migratory stream (RMS), granule cell layer (GCL), mitral cell layer (MCL), external plexiform layer (EPL), glomerular layer (GL) and olfactory nerve layer (ONL) (Fig. 3c–e). Moreover, STAligner identified two slice-specific spatial structures, the accessory olfactory bulb (AOB) and the granular layer of the accessory olfactory bulb (AOBgr) in the Slide-seqV2 ST slice, which may be caused by different sampling locations (Fig. 3c–e). The UMAP visualization of STAligner embeddings showed that the slice-shared laminar structures were mixed and the slice-specific structures were separated. However, Harmony and SEDR cannot recover these two structures and reveal such differences between these two ST slices (Supplementary Fig. 7). Furthermore, these spatial domains identified by STAligner were characterized by known marker genes (Fig. 3e). For example, the expression of *Fxyd6*²² showed a distinct spatial distribution on the recognized AOB domain, and the expression of the granular cell marker *Atp2b4*²³ showed a distinct spatial distribution on the recognized AOBgr domain. The narrow loop structure of the MCL was enriched in the spatial expression of mitral cell marker *Gabra1*²⁴. These results demonstrate that STAligner can integrate slices with different tissue structure compositions and avoid overcorrection of the slice-specific spatial domains.

Identifying developmental mouse embryo tissue structures

We applied STAligner to integrate four mouse embryo slices sampled at the time stages of E9.5, E10.5, E11.5 and E12.5 profiled by Stereo-seq⁶ to investigate the spatiotemporal heterogeneity of tissue structures during mouse organogenesis. Although the four slices have very different sizes and there are obvious batch effects, STAligner can effectively integrate the four slices in the embedding space and align the spatial domains, which showed very consistent localization spatially across the four embryonic stages (Fig. 4a and Supplementary Fig. 8a). Each domain showed clear tissue structure and can be annotated based on the anatomic location using eHistology Kaufman annotations²⁵ (Fig. 4a). To reconstruct the developmental trajectory of each tissue structure, we built connections between the same spatial domains in successive stages and generated an acyclic directed graph to reflect the ancestor–descendant relationships (Fig. 4b). According to the origination from the three germ layers during gastrulation, the annotated organs and tissues can be summarized into three parts, that is, ectoderm, mesoderm and endoderm. Notably, the tissues originated from different germ layers also showed clear co-localization patterns based on the UMAP visualization (Supplementary Fig. 8a). More importantly, the annotated tissues across the four stages were characterized with some known marker genes⁶, such as *Sox2* for the forebrain, *Ina* for the hindbrain, *Tbr1* for the olfactory bulb, *Six6* for the hypothalamus, *Myl7* for the heart, *Trff* for the liver, *Pax1* for the cartilage primordium and *Myog* for the muscle (Fig. 4c and Supplementary Fig. 8b).

All the major tissues and organs (for example, heart, liver and brain) were detected and consistent with the well-known anatomical structure of the organ (Fig. 4b,c). In particular, the heart was accurately identified with nearly unchanged proportions over the course of the developmental period, which confirmed the fact that the heart is the first organ formed and functioned in the early embryo²⁶. The liver

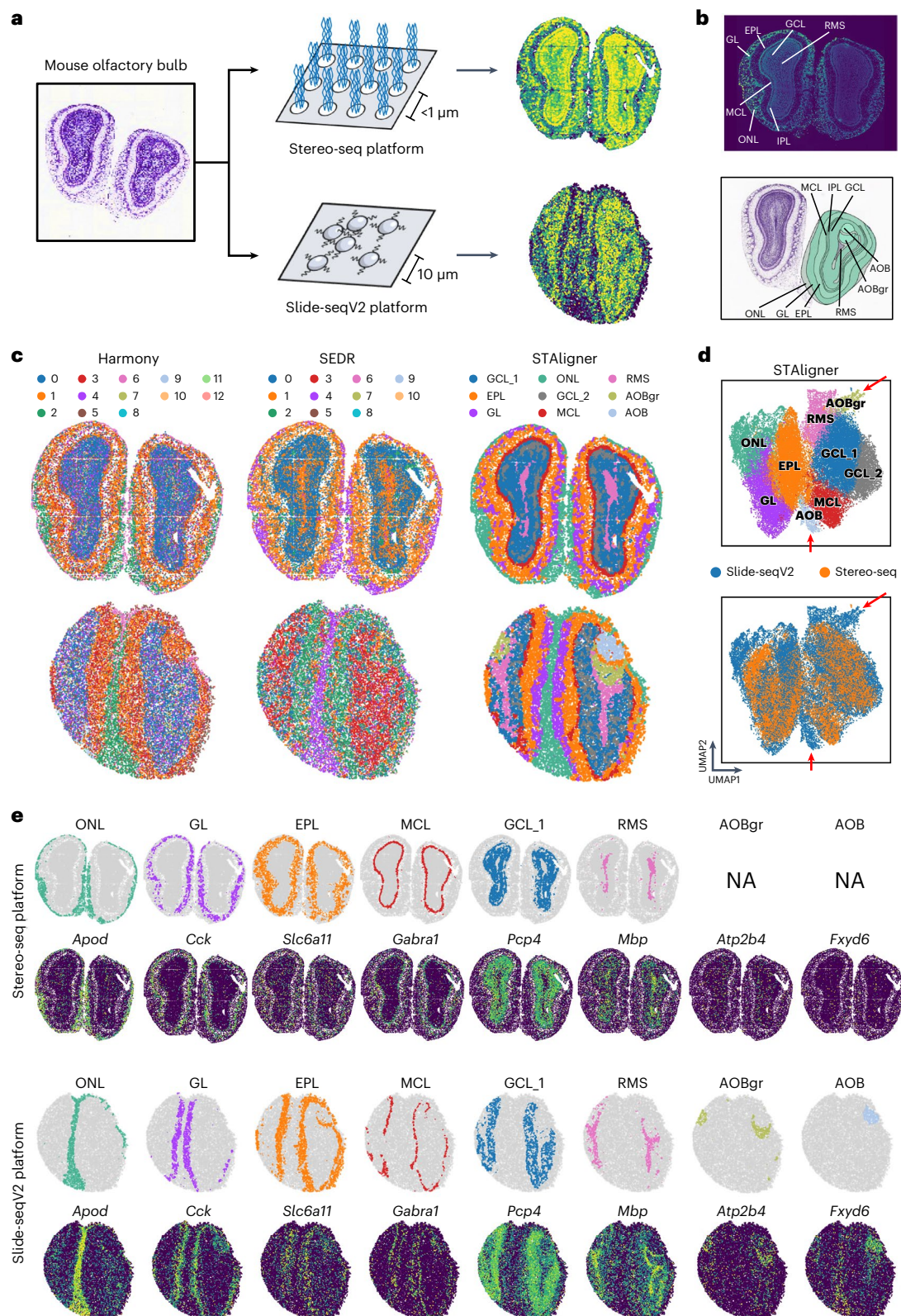


Fig. 3 | Identifying common and unique tissue structures in the mouse olfactory bulb data across two different sequencing platforms (Stereo-seq and Slide-seqV2). **a**, Two mouse olfactory bulb datasets produced by Stereo-seq and Slide-seqV2. **b**, Laminar organization of mouse olfactory bulb annotated by DAPI-stained image (top) and the Allen Mouse Brain Atlas (bottom). Internal plexiform layer (IPL). **c**, Visualization of spatial domains identified simultaneously

by integrating and aligning the two ST slices profiled via Harmony, SEDR and STAligner, respectively. **d**, UMAP plots of STAligner embeddings colored by spatial domain (top) and slices (bottom). **e**, Spatial domains identified simultaneously by STAligner and the corresponding marker genes in the slice of Stereo-seq (top) and Slide-seqV2 (bottom). NA, missing structures.

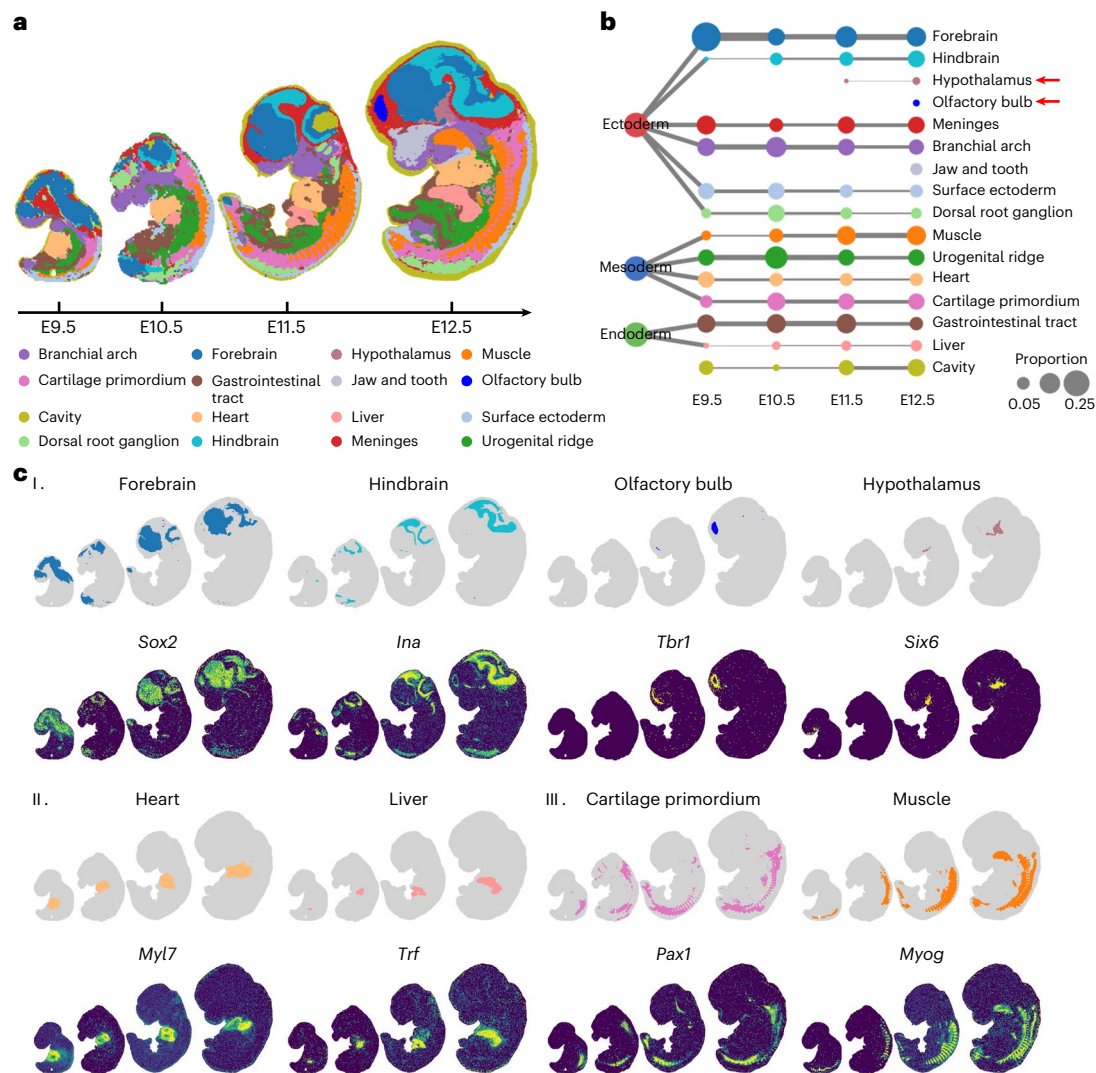


Fig. 4 | Revealing developmental dynamics of anatomical tissue substructures (spatial domains) in the early mouse embryo data. **a**, Aligned spatial domains characterized simultaneously by integrating and aligning four ST data of four embryonic slices (E9.5 to E12.5) via STAligner. **b**, Directed acyclic graph showing the developmental dynamics of tissue substructures. Each row

corresponds to one of the tissue substructures (spatial domains) shown in **a**. **c**, Spatial tissue substructures (top) relating to the brain (I), heart and liver (II), and cartilage primordium and muscle (III), and their corresponding marker genes (bottom).

occupied a small area at the earliest stage E9.5 and rapidly increased size in the following stages. Likewise, we observed prominent increase in the proportions of hindbrain and muscle, indicating the proliferation of these tissue structures during the four stages of organogenesis. In addition, STAligner can also delineate tissues with special shapes, such as the meninges with a continuous closed loop at late embryonic stages. STAligner can also detect small (proportion <5%) structures, such as the hypothalamus and olfactory bulb, at later developmental stages (after E11.5), which were supported by known histological evidence. The hypothalamic neurogenesis starts at E11.5 and reaches the peak at E12.5²⁷ and the development of olfactory bulb initiated at E10.5 and cannot be detected macroscopically until E12.5^{28,29}. These results demonstrate that STAligner can integrate the ST slices of different developmental stages, facilitate the cross-stage comparison and deepen our understanding on the developmental dynamics of the anatomical structure changes during mouse organogenesis.

With the accumulation of ST data in both human and mouse, it is interesting and useful to identify consensus regions between them, making translational studies easier. We compared the embryo slice in mouse (at E10.5) and human (at Carnegie stage (CS) 13) profiled

by Stereo-seq and 10x Visium technology, respectively^{6,30}. We used a one-to-one homologous relationship and applied STAligner to these two slides directly. It is interesting to see that most tissue structures between them were conserved (Supplementary Fig. 9). We also found a mouse-specific region dermomyotome, which may be attributed to the difference in developmental speeds between mouse and human organogenesis.

Identifying disease-related spatial domains

Here we applied STAligner to two ST slices from normal⁵ and Alzheimer's disease (AD)⁸ mouse hippocampus tissue profiled by Slide-seqV2, and identified 16 distinct spatial domains. Most spatial domains revealed clear spatial patterns and corresponded to anatomical structures of hippocampus in the Allen Mouse Brain Atlas (Fig. 5a,b and Supplementary Fig. 10a). The UMAP visualization of STAligner embeddings showed that the common domains from the two slices were highly overlapping and slice-specific domains were separated (Fig. 5c). STAligner could capture shared tissue structures between these two conditions. For example, domains 4, 7 and 11 constructed a clear 'cord-arrow-like' structure in the hippocampal region in both the two slices (Fig. 5d and Supplementary

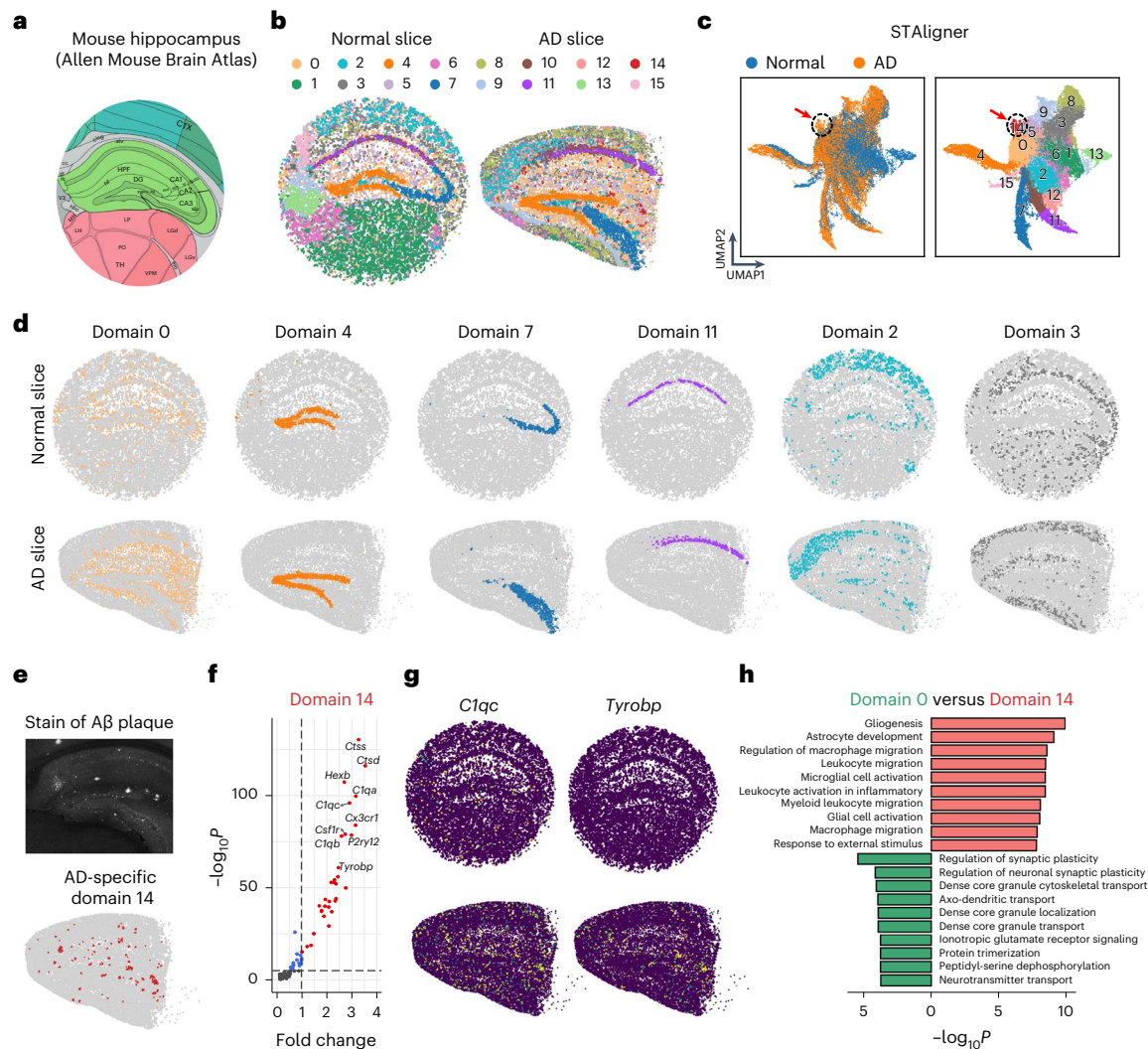


Fig. 5 | Integrating two mouse hippocampus ST slices of normal and AD conditions. **a**, Laminar organization of mouse hippocampus annotated by the Allen Mouse Brain Atlas. **b**, Visualization of aligned spatial domains characterized simultaneously by STAligner between the normal and AD mouse hippocampus datasets produced by Slide-seqV2. **c**, UMAP plots of STAligner embeddings colored by condition types and spatial domains. **d**, Visualization of spatial domains identified simultaneously between the two datasets. **e**, The spatial

distribution of spatial domain dominantly found in the AD slice (bottom) and the antibody stain of A β plaque in the adjacent hippocampus slice (top). **f**, Volcano plot of DEGs between domain 14 and domain 0. **g**, Spatial expression maps of two marker genes *C1qc* and *Tyrobp* for spatial domain 14 in the normal and AD slices. **h**, GO enrichment terms of the DEGs. Panel **e** adapted with permission from ref. 8, Springer Nature America, Inc.

Fig. 10a). According to the Allen Mouse Brain Atlas, domain 4 ('arrow-like') corresponds to the dentate gyrus region, and domains 7 and 11 ('cord-like') correspond to the structures CA1 and CA3 of Ammon's horn, respectively, which were supported by some known gene markers. For example, *Lrrtm4* regulating excitatory synapse development and AMPA receptor trafficking in dentate gyrus granule cells was expressed in the dentate gyrus field³¹.

To find the differences between AD and normal tissues, we first computed the composition of each domain belonging to the AD or normal slice (Supplementary Fig. 10b). We found that domains 0 and 14 were located at the central hippocampal region (except for the 'cord-arrow-like' structure), but showed clear different compositions between slices, where domain 14 was more enriched in the AD slice. From the UMAP visualization, we can observe that domain 14 showed a distinct position and most spots of it belonged to the AD slice, although it was close to domain 0 (Fig. 5c). Importantly, the high A β plaque density in the vicinity of domain 14 indicated that it was highly colocalized with the amyloid deposition, which is one of the key pathological features of AD (Fig. 5e and Supplementary Fig. 10c). These

observations suggested that domain 14 was AD associated. Moreover, the highly expressed genes (significantly upregulated) in domain 14 compared with domain 0 included microglia marker genes and some of them were implicated with the AD risk (Fig. 5f). For instance, *Csf1r* is a key growth factor receptor for microglia and has a crucial role for its homeostasis and development³²; and *Tyrobp* was found to be markedly upregulated in A β -associated brain regions³³ and could enhance phagocytic activity of microglia and clear A β deposits³⁴. An earlier study demonstrated that *C1qa*, *C1qb* and *C1qc* are strongly expressed for A β -associated microglia and could mediate synapse loss in AD³⁵. We also find that the spatial expression distribution of differentially expressed genes (DEGs; *C1qc* and *Tyrobp*) is localized near the position of A β accumulation, illustrating the potential effect of A β accumulation on gene expression (Fig. 5g). These DEGs were involved in many Gene Ontology (GO) terms such as gliogenesis, astrocyte development and microglial cell activation (Fig. 5h). These results demonstrate that STAligner could preserve true biological variations while removing batch effects, and untangle the dysregulated cellular network surrounding the A β deposition of AD.

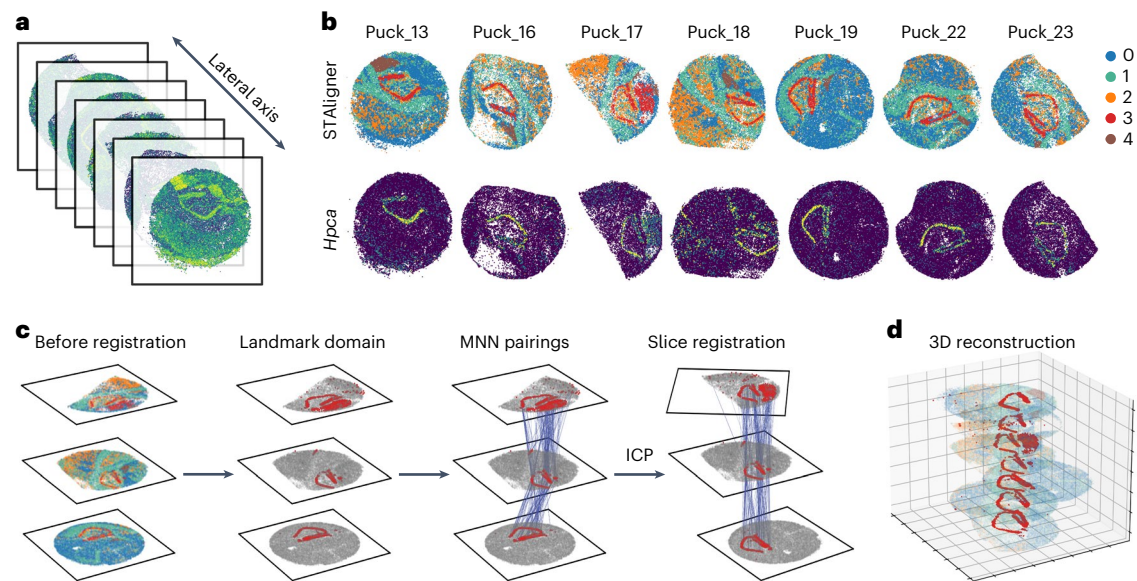


Fig. 6 | Stacked 3D alignment of adjacent mouse hippocampus slices. a, Seven mouse hippocampus slices produced by Slide-seq. Spots in the slice are colored by the number of transcripts. **b**, Visualization of spatial domains identified simultaneously by integrating seven consecutive slices (top) via by STAligner and the expression of marker gene *Hpca* for domain 3 (bottom). **c**, The workflow

of downstream registration including landmark domain extraction shared by adjacent slices, MNN pair search between adjacent slices as anchor pairs and pairwise registration using the ICP algorithm. **d**, Registered slices with the hippocampus structure highlighted.

To demonstrate the performance of STAligner on heterogeneous tissues, we also applied it to three human breast cancer slices from different patients sequenced by the 10x Visium platform (<https://www.10xgenomics.com/resources/datasets>). We found that the identified domains were highly consistent with histological annotation regions of invasive carcinoma (domains 4, 5, 6, 8, 10), carcinoma in situ (domains 7, 9, 11) and non-tumor (domains 1, 2, 3) (Supplementary Fig. 11a,b). Moreover, domains 4, 10 and 11 were more slice specific, which may be due to the heterogeneity among patients with tumors. Further differential expression analysis found that domains 4 and 10 exhibited a strong spatial expression pattern of genes *VTCN1* and *NRAS*, respectively (Supplementary Fig. 11c). *VTCN1* was recognized as immunosuppressive gene³⁶, indicating that domain 4 could be a region of immunosuppressive status and has high potential for cancer invasion. It has been reported that *NRAS* can mediate autocrine/paracrine signaling and promote cell invasiveness in breast cancer³⁷. We also found increased expression of the gene *FCGR3B* for domain 11. *FCGR3B* is a classical marker gene for natural killer cells or neutrophils³⁸, indicating that domain 11 could be a region of cancer growth limited by immune activities. These results shown that STAligner can delineate the difference in tumor substructures across tumor samples.

Three-dimensional reconstruction of consecutive mouse hippocampus slices

For the same tissue, multiple consecutive adjacent tissue slices need to be manually registered by rigid rotation and translation for 3D tissue reconstruction. Here we demonstrated the capability of automatic registration of seven adjacent mouse hippocampus slices generated by Slide-seqV1⁴ (Fig. 6a). It is challenging to integrate and register these slices due to the low transcript detection sensitivity of Slide-seqV1 and only partial overlap among these adjacent slices. STAligner could successfully integrate the seven mouse hippocampus slices, and recognize the shared structure of corpus callosum (domain 1), Ammon's horn and the dentate gyrus (domain 3), and third ventricle (domain 4), which demonstrated high consistency with the hippocampus annotations in the Allen Mouse Brain Atlas (Figs. 5a and 6b). The UMAP visualization of the STAligner embeddings demonstrated that the spots across slices

were mixed while the spatial domains were separated (Supplementary Fig. 12a) with a clear expression pattern (Supplementary Fig. 12b). These tissue structures were further confirmed by the higher expressions of oligodendrocytes marker *Plp1*³⁹ (corpus callosum), *Hpca*⁴⁰ (Ammon's horn and the dentate gyrus) and choroid plexus marker *Ttr*⁴¹ (third ventricle) (Supplementary Fig. 12c). Due to the difference between sampling position in the experimental procedures, the tissues in Puck_16–18, 22–23 were incomplete, resulting in the differences of the shape and distribution of the tissue structures across the seven tissue slices. However, embedding alignment does not mean geometrical registration. According to the orientation and position of domains shared across slices (domain 3), we can observe that the adjacent slices were misaligned (Fig. 6b, bottom panel).

Intuitively, using a common domain with asymmetrical shape and enough overlapping areas could make the further registration process robust against noise. We used the domain 3 as a landmark domain, which has a clear 'cord-arrow-like' structure, has major role in memory and spatial navigation, and is a major site of adult neurogenesis^{42,43}. The MNNs in this domain identified by STAligner were selected as anchor pairs to guide the registration process using the Iterative Closest Point (ICP) algorithm (Fig. 6c). All adjacent slices were aligned sequentially with translation and rotation along the z axis. From the stacked 3D tissue with registered coordinates, we can observe that the orientation and position of the 'cord-arrow-like' structure were matched accordingly (Fig. 6d), resulting in a clear visual improvement of 3D spatial patterns. By contrast, PASTE, which was designed for 3D reconstruction of globally similar ST slices, cannot correctly align these slices with partial overlap and thus failed to reconstruct the 3D 'cord-arrow-like' structure (Supplementary Fig. 13). These results illustrate that the MNNs and spatial domains generated by STAligner could help to guide the coordinate registration of adjacent slices and achieve promising 3D tissue reconstruction for diverse tissue slices.

Evaluating the computational time and memory usage

The rapid development of ST technologies results in higher spatial resolution, larger field of view and larger number of spots. These large-scale datasets require the integration method to be memory

and time efficient. We recorded the computational time and graphics processing unit (GPU) memory usage cost by STAligner on three datasets in Supplementary Fig. 14a. When integrating the largest mouse hippocampus dataset with seven slices and more than 170,000 spots in total, STAligner is fast and takes only 8.5 min. We can also find that the GPU memory usage remains constant when increasing the number of slices (Supplementary Fig. 14b), which makes STAligner suitable for analyzing large-scale ST datasets.

Discussion

STAligner aims to identify consensus spatial domains across slices, which is different from the goal of the non-integration methods Bayes-Space¹¹, SpaGCN¹⁰ and STAGATE¹². These tools were mainly designed for spatial domain identification in each ST slice and cannot be directly used for simultaneous domain identification across slices. Spatial domain identification for a single slice will only reveal them separately without the domain alignment. They have to rely on some downstream analysis to determine such correspondence. The integration could facilitate the multi-slice ST analysis and identify consensus domains missed by the non-integration methods.

Currently, our 3D reconstruction strategy based on the ICP algorithm is limited to rigid transformation and cannot account for local flexible distortions. Future work is expected to develop a spatial-domain-guided flexible registration method for 3D reconstruction. The flexible transformation strategy can even be used to register slices from different samples while handling the anatomical differences across samples, build a common reference of organs of different individuals and construct ST atlases in the future⁴⁴.

In addition, the current STAligner extracts only gene expression and spatial information. The model can be further extended to integrate multimodal data, such as histological images and epigenomic data. We anticipate that future advancements in these directions would facilitate more comprehensive exploration of biological phenomena.

Methods

Data preprocessing

STAligner takes gene expression and spatial coordinate information of multiple ST slices as input. For each slice, we normalize the raw gene expressions according to library size and log-transformed using the SCANPY package⁴⁵. Then, we select the top-5,000 highly variable genes. We take the intersection of the highly variable genes of all the input slices to ensure the spot features are shared across slices.

Construction of the spatial neighbor graph

On the basis of the spatial coordinate information of all spots, we measure the similarities between spots by the Euclidean distance. Two spots are considered as neighbor if the distance between them is shorter than a pre-defined threshold r . Using the spatial neighbor relationship, we construct an undirected neighbor graph denoted as the adjacency matrix A , where $A_{ij} = 1$ if and only if there is an edge between spot i and spot j . A is a symmetric matrix with self-loops added for each spot. We can adjust r so that each spot has 4–15 nearest neighbors on average for diverse ST scenarios and platforms.

Spatially aware spot triplet learning

STAligner consists of two parts: the graph attention autoencoder for spatially aware embedding learning and the spot triplet learning for batch correction (Fig. 1).

Graph attention autoencoder. The graph attention autoencoder⁴⁶ is an unsupervised representation learning framework using both graph structure information and node attributes. It employs the graph attention mechanism to adaptively determine the influence of spatial neighbor nodes on learning node representations and reduce the effect of the noise and missingness in the expression data. Specifically, the graph

attention encoder takes the normalized gene expressions X and the corresponding graph structure as input, to get spot representations $\mathbf{z}$. Let L be the number of layers in the encoder and N be the number of spots in the slices. The output representation of the k th ($k \in \{1, 2, \dots, L-1\}$) encoder layer of spot $i \in \{1, 2, \dots, N\}$ is formulated as follows:

$$\mathbf{z}_i^{(k)} = \sigma \left(\sum_{j \in \mathcal{N}_i} \alpha_{ij}^{(k)} \left(W^{(k)} \mathbf{z}_j^{(k-1)} \right) \right), \quad (1)$$

where $\mathcal{N}_i$ represents the neighbor set of spot i (including spot i itself), σ is the nonlinear activation function, $W^{(k)}$ is the trainable weight matrix, $\mathbf{z}_i^{(0)}$ is the initial gene expression $\mathbf{x}_i$ of spot i . $\alpha_{ij}^{(k)}$ is set to make the relevance coefficients of spot i 's neighbors comparable:

$$\alpha_{ij}^{(k)} = \frac{\exp(e_{ij}^{(k)})}{\sum_{l \in \mathcal{N}_i} \exp(e_{il}^{(k)})}, \quad (2)$$

where $e_{ij}^{(k)}$ indicates the relevance of spot i to its neighboring spot j , and can be computed by:

$$e_{ij}^{(k)} = \text{Sigmoid} \left(\mathbf{v}^{s(k)T} \left(W^{(k)} \mathbf{z}_i^{(k-1)} \right) + \mathbf{v}^{r(k)T} \left(W^{(k)} \mathbf{z}_j^{(k-1)} \right) \right), \quad (3)$$

where $\mathbf{v}^{s(k)}$ and $\mathbf{v}^{r(k)}$ are the trainable weight parameters of the k th layer, and Sigmoid denotes the sigmoid activation function. The L th encoder layer does not use the attention mechanism and can be calculated by:

$$\mathbf{z}_i^{(L)} = \sigma \left(W^{(L)} \mathbf{z}_i^{(L-1)} \right), \quad (4)$$

which is considered as the final embedding of spot i .

The number of layers of the decoder is the same as the encoder. To reconstruct expression features, the decoder reverses the process of learning latent embedding in the encoder. Using the output of the encoder as the input of the decoder, the k th ($k \in \{2, \dots, L-1, L\}$) decoder layer reconstructs the representation of spot i in layer $k-1$ as follows:

$$\hat{\mathbf{z}}_i^{(k-1)} = \sum_{j \in \mathcal{N}_i} \hat{\alpha}_{ij}^{(k-1)} \sigma \left(\hat{W}^{(k)} \hat{\mathbf{z}}_j^{(k)} \right). \quad (5)$$

Similar to the encoder, the attention mechanism is not used in the last layer:

$$\hat{\mathbf{z}}_i^{(0)} = \sigma \left(\hat{W}^{(1)} \hat{\mathbf{z}}_i^{(1)} \right). \quad (6)$$

The output $\hat{\mathbf{z}}_i^{(0)}$ is considered as the reconstructed expressions. To avoid overfitting, STAligner sets $\hat{W}^{(k)} = W^{(k)T}$ and $\hat{\alpha}_{ij}^{(k)} = \alpha_{ij}^{(k)}$ respectively.

The objective function of the graph attention autoencoder is to maximize the similarity between input and reconstructed expressions, achieved by minimizing the mean square error loss:

$$L_{\text{rec}} = \frac{1}{N} \sum_{i=1}^N \|\hat{\mathbf{z}}_i^{(0)} - \mathbf{x}_i\|_2. \quad (7)$$

Spot triplet learning. We aim to enforce the encoder to learn a batch-corrected embedding for multiple ST data via the triplet learning, which has shown superior performance in diverse fields including scRNA-seq dataset integration⁴⁷. We define the spot triplet consisting of an anchor, a positive spot and a negative spot. It should be pointed out that, to avoid potential batch effects and dropouts, the triplets are defined in the embedding space instead of original gene expression space. The formed anchor–positive pair is used to overcome the batch effect while the anchor–negative pair guides the alignment in a discriminative way. We use the MNN pairs to define the anchor and positive spot that have similar gene expression but are from two different slices. The negative spot is randomly sampled from the same

slice as the anchor spot. As the total number of spots is usually much larger than the k -nearest neighbors of the anchor spot, the probability of the sampled negative spot locating in the k -nearest neighbors of the anchor is very low. This negative sampling strategy can accelerate the model training, especially in large-scale datasets. The triplet loss is used to minimize the distance between the anchor–positive pair and maximize the distance between the anchor–negative pair in the latent space $\mathbf{z}$. It is formulated with a Euclidean distance function as follows

$$L_{\text{tri}} = \frac{1}{N_{\text{tri}}} \sum_{(a,p,n) \in \mathcal{C}_{\text{tri}}} \max(\|\mathbf{z}_a^{(L)} - \mathbf{z}_p^{(L)}\|_2 - \|\mathbf{z}_a^{(L)} - \mathbf{z}_n^{(L)}\|_2 + \tau, 0), \quad (8)$$

where a , p and n represent the indices of the anchor, positive and negative spots respectively, $\mathcal{C}_{\text{tri}}$ is the set of the identified spot triplets with size N_{tri} and τ (default 1.0) is the margin used to enforce the distance between positive and negative pairs. Finally, the final objective is written as:

$$L = (1 - \alpha)L_{\text{rec}} + \alpha L_{\text{tri}}. \quad (9)$$

where α is a hyperparameter that control the contribution of L_{rec} and L_{tri} in the loss function, and is set by 0.5 by default. The effect of hyperparameter α on performance is shown in Supplementary Fig. 15b.

We perform spot triplet detection and graph attention autoencoder training in an iterative manner to generate refined triplets and improved embeddings. We first detect spot triplets in the embedding space generated by the encoder from the last iteration, and then guide the embedding alignment process with the detected triplets. Here, the initial embeddings are generated without using the triplet loss.

Landmark spatial domain guided consecutive slice registration and 3D tissue reconstruction

Registration of the spot coordinates of the adjacent slices is an important step for 3D tissue reconstruction. A typical registration process uses landmark points to guide the registration, including three steps: landmark point (key point) extraction, landmark point matching and the optimal transformation estimation. The ICP algorithm^{48,49} is a typical method to achieve rigid transformation (that is, translation and rotation) to register the coordinates from different coordinate systems to a common coordinate framework with given landmark point matching. However, it is challenging to directly apply the algorithm to adjacent slices due to partial overlapping and dropout noise. Instead of detecting the correspondences for each spot in the adjacent slices, we consider the MNN spots as landmark points that come from a selected landmark domain shared across slices. Here, the landmark domain should be asymmetrical and have enough overlapping areas to improve the robustness of the corresponding points. Using the coordinates of the MNN spots as the input of the ICP algorithm, we can obtain a transformation matrix that can be further used to map all the spots of the target to the reference slice. The registration process is performed in a sequential pairwise manner across all the adjacent ST slices.

The overall architecture of STAligner

The encoder of STAligner consists of two layers with node dimensions set as 512 and 30, respectively, and the decoder is just the reverse of the encoder. The Adam optimizer⁵⁰ is used to minimize the loss function with learning rate of 10^{-4} . We set the weight decay as 10^{-4} and the activation function as the exponential linear unit⁵¹. We adopt the PyTorch Geometric library⁵² to implement the graph attention autoencoder. We train the model in two steps: the graph autoencoder model is first trained without the triplet loss (500 iterations by default) and generates the embeddings for triplet identification. Then, we alternate the model training and triplet refinement every 100 iterations until the fixed number of iterations reached (500 iterations by default).

All hyperparameters were optimized based on the performance on the DLPFC data (Supplementary Fig. 15a).

Clustering

The spatial domains were identified by clustering algorithm with the embeddings from STAligner as input. For datasets with prior knowledge about the number of domain labels, the mclust clustering algorithm implemented by R package mclust (v5.4.9) was used⁵³. For datasets without prior knowledge, the Louvain algorithm implemented by SCANPY package (v1.9.1) was employed. The ARI is used to quantify the agreement between the clustering result and the original annotation. A higher ARI score means better agreement between two grouping results while a zero score means that the clustered labels are randomly guessed.

Differential expression analysis and GO enrichment analysis

We used the Wilcoxon test implemented in the FindMarkers() function of the R package Seurat (v4.1.0) to identify DEGs for the spatial domain in normal and disease slices with a cut-off of an adjusted P value (Benjamin–Hochberg correction) of 0.01. We performed the GO enrichment analysis for the domain-specific DEGs using the R package clusterProfiler (v4.2.2)⁵⁴.

Benchmarking integration methods

To benchmark the integration performance, we considered the following integration methods: Harmony¹⁷, SEDR⁹ and PASTE¹⁹. We also used SCANPY⁴⁵ to produce uncorrected embeddings of the combined raw input data, where principal component analysis (PCA) was used to perform dimensionality reduction and no integration method was used. All parameters were kept at default values.

Harmony. The high performance of Harmony has been shown in a previous benchmarking study for scRNA-seq data integration⁵⁵. According to the standard workflow, PCA was first performed on the combined expression matrix, and then the obtained PCA embeddings and batch information were considered as input of the scanpy.external.pp.harmony_integrate() function in the SCANPY package. Finally, the corrected PCA embeddings were obtained as output. The number of principal components is set as 50 (default setting).

SEDR. SEDR is a spatial embedding representation method based on graph neural network with PCA embeddings as node features. In the original study, the authors fed the latent embeddings of SEDR to Harmony for batch correction. We used the demo code run_UBC_DLPFC_data.py in the online tutorial for unsupervised batch correction experiments. We empirically disabled the deep embedded clustering (DEC) loss function to obtain better performance. The number of principal components is set as 300 and the epoch is set as 300 (default setting).

PASTE. Using optimal transport that leverages both transcriptional similarity and spatial distances between spots, PASTE integrates multiple adjacent slices from the same sample in two alignment modes. The users can choose the mode by the biological problem of interest. The first mode finds an optimal probabilistic mapping between spots in two adjacent slices and uses the mapping to achieve spatial coordinate registration. By pairwise alignment between multiple pairs of adjacent slices, a stacked 3D spatial reconstruction can be obtained. We ran this mode on the seven mouse hippocampus slices (Supplementary Fig. 13). The second mode infers an enhanced ‘consensus’ slice by integrating multiple adjacent slices that can increase the power of downstream analysis. The low-dimensional embeddings in the integrated consensus slice can be used for further clustering analysis. We ran this mode on each DLPFC slice and concatenated the embeddings of each slice for evaluation. We ran PASTE according to the online tutorial

getting-started.ipynb where the `pst.pairwise_align()` function and `pst.center_align()` function were used for the first and second modes, respectively. In the second mode, the size of low-dimensional representation is set as 15 (default setting).

Data description

The 10x Visium human DLPFC dataset was collected from three independent adult samples. Each sample consists of four adjacent tissue slices. The manually annotated labels in the original study include white matter and six neocortical layers. We used these annotations as ground truth to evaluate the performance of simultaneous spatial domain identification and the quality of batch-corrected embeddings. Each slice contains ~4,000 spots, ranging from 3,498 to 4,789. The mouse brain dataset was profiled by 10x Visium platform and the tissue was divided into two slices in the sagittal direction. The anterior and posterior slice contains 2,695 and 33,55 spots, respectively. The adult mouse olfactory bulb dataset was profiled by Stereo-seq and Slide-seqV2, respectively. The Stereo-seq slice processed by an early study⁹ contains 19,109 spots. The Slide-seqV2 slice contains 20,139 spots. The early mouse embryo dataset consisting of four stages (E9.5, E10.5, E11.5 and E12.5) profiled by Stereo-seq, contains 5,913, 18,408, 30,124 and 51,365 spots, respectively. The ST data of normal and AD disease mouse hippocampus slices both profiled by Slide-seqV2 contain 19,285 and 15,092 spots, respectively. For the AD slice, an adjacent A β plaque staining image of the same brain tissue was also collected. The mouse hippocampus data consisting of seven adjacent slices profiled by Slide-seqV1 with the number of spots ranging from 14,860 to 32,895. Slide-seqV1 data are highly noisy and sparse with lower transcript detection sensitivity than Slide-seqV2⁵. Thus, recovering the true biology patterns is more challenging. The three human breast cancer slices based on 10x Visium platform consist of 4,727, 3,798 and 4,992 spots. The human embryo slice was profiled by 10x Visium platform, containing 2,909 spots. A summary of these datasets can be found in Supplementary Table 1.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Source data for Figs. 2–6 is available with this paper. The datasets analyzed in this study are all from publicly available datasets (Supplementary Table 1). Specifically, the human DLPFC dataset can be accessed in the spatialLIBD package (<http://spatial.libd.org/spatial-LIBD>). The mouse olfactory bulb tissue data generated by Stereo-seq and Slide-seqV2 platforms can be accessed from https://github.com/JinmiaoChenLab/SEDR_analyses and https://singlecell.broadinstitute.org/single_cell/study/SCP815, respectively. The mouse sagittal posterior and anterior brain data can be accessed at https://support.10xgenomics.com/spatial-gene-expression/datasets/1.0.0/V1_Mouse_Brain_Sagittal_Posterior and https://support.10xgenomics.com/spatial-gene-expression/datasets/1.0.0/V1_Mouse_Brain_Sagittal_Anterior, respectively. The mouse embryo data can be accessed at <https://db.cngb.org/stomics/mosta/>. The human embryo data can be accessed at <https://heoa.shinyapps.io/code/>. The human breast cancer data can be accessed at https://cf.10xgenomics.com/samples/spatial-exp/1.0.0/V1_Breast_Cancer_Block_A_Section_1/V1_Breast_Cancer_Block_A_Section_1_web_summary.html, https://cf.10xgenomics.com/samples/spatial-exp/1.2.0/V1_Human_Invasive_Ductal_Carcinoma/V1_Human_Invasive_Ductal_Carcinoma_web_summary.html and https://cf.10xgenomics.com/samples/spatial-exp/2.0.0/CytAssist_FFPE_Human_Breast_Cancer/CytAssist_FFPE_Human_Breast_Cancer_web_summary.html. The normal and Alzheimer's disease mouse hippocampus data can be accessed at https://singlecell.broadinstitute.org/single_cell/study/SCP815 and https://singlecell.broadinstitute.org/single_cell/study/SCP1663, respectively. The Slide-seq 3D mouse

hippocampus slice can be accessed at https://singlecell.broadinstitute.org/single_cell/study/SCP354/slide-seq-study. The annotation images from the Allen Mouse Brain Atlas can be accessed at <https://mouse.brain-map.org/static/atlas> (Supplementary Table 2).

Code availability

An open-source Python implementation of the STAligner package is available at <https://github.com/zhanglabtools/STAligner> and <https://doi.org/10.5281/zenodo.8315415> ref. 56.

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Author contributions

S.Z. conceived and supervised the project. X.Z. developed and implemented the STAligner algorithm. X.Z., K.D. and S.Z. validated the methods and wrote the paper. All authors read and approved the final paper.

Competing interests

The authors declare no competing interests.

Additional information

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Data collection	All data analyzed within this manuscript are publicly available. No additional software was used for the data collection process.
Data analysis	We performed the data analysis using our developed STAligner software (Version: 1.0.0) in this paper, which is available at GitHub repository: https://github.com/zhanglabtools/STAligner . The following software packages were used in our code: python v3.8.0, scanpy v1.9.1, torch v1.9.1, torch-geometric v2.0.3, numpy v1.20.3, anndata v0.7.8, pandas v1.2.3, scipy v1.5.3, matplotlib v3.5.1, louvain v0.7.1, umap-learn v0.5.2, rpy2 v2.9.5, seaborn v0.11.2, Seurat v4.2.1, PASTE v1.4.0, SEDR v1.0.0, harmony v1.2.3360, mclust v5.4.9, clusterProfiler v4.2.2.

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- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

Source data for Figures 2-6 is available with this manuscript. The datasets analyzed in this study are all from publicly available datasets. Specifically, the human

DLPFC dataset can be accessed in the spatialLIBD package (<http://spatial.libd.org/spatialLIBD>). The mouse olfactory bulb tissue data generated by Stereo-seq and Slide-seqV2 platforms can be accessed from https://github.com/JinmiaoChenLab/SEDR_analyses, https://singlecell.broadinstitute.org/single_cell/study/SCP815, respectively. The mouse sagittal posterior and anterior brain data can be accessed at https://support.10xgenomics.com/spatial-gene-expression/datasets/1.0.0/V1_Mouse_Brain_Sagittal_Posterior; https://support.10xgenomics.com/spatial-gene-expression/datasets/1.0.0/V1_Mouse_Brain_Sagittal_Anterior. The mouse embryo data can be accessed at <https://db.cngb.org/stomics/mosta/>. The human embryo data can be accessed at <https://heoa.shinyapps.io/code/>. The human breast cancer data can be accessed at https://cf.10xgenomics.com/samples/spatial-exp/1.0.0/V1_Breast_Cancer_Block_A_Section_1/V1_Breast_Cancer_Block_A_Section_1_web_summary.html, https://cf.10xgenomics.com/samples/spatial-exp/1.2.0/V1_Human_Invasive_Ductal_Carcinoma/V1_Human_Invasive_Ductal_Carcinoma_web_summary.html, and https://cf.10xgenomics.com/samples/spatial-exp/2.0.0/CytAssist_FFPE_Human_Breast_Cancer/CytAssist_FFPE_Human_Breast_Cancer_web_summary.html. The normal and Alzheimer's disease mouse hippocampus data can be accessed at https://singlecell.broadinstitute.org/single_cell/study/SCP815 and https://singlecell.broadinstitute.org/single_cell/study/SCP1663, respectively. The Slide-seq 3D mouse hippocampus slice can be accessed at https://singlecell.broadinstitute.org/single_cell/study/SCP354/slide-seq-study. The annotation images from the Allen Mouse Brain Atlas can be accessed at <https://mouse.brain-map.org/static/atlas>.

Human research participants

Policy information about [studies involving human research participants and Sex and Gender in Research](#).

Reporting on sex and gender

N/A

Population characteristics

N/A

Recruitment

N/A

Ethics oversight

N/A

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see [nature.com/documents/nr-reporting-summary-flat.pdf](https://www.nature.com/documents/nr-reporting-summary-flat.pdf)

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size

No sample size calculation was performed. All data used in this study were obtained from public resources and used to demonstrate the ability and functionality of STAligner. The DLPFC dataset contains twelve slices collected from three adult samples with known spot labels and were determined to quantitatively evaluate the performance in integrating multiple samples. The anterior and posterior slice collected from mouse brain in the sagittal direction were determined to stitch adjacent slices into a larger one for deciphering global tissue structure. Two mouse olfactory bulb slices profiled by Slide-seqV2 and Stereo-seq platforms were determined to evaluate the performance in integrating slices from different platforms. Four mouse embryo slices were determined to demonstrate the ability in integrating slices from different developmental time stages. Two normal and Alzheimer's mouse hippocampus slices were determined to demonstrate the ability in integrating slices from different disease conditions. Seven adjacent mouse hippocampus slices were determined to demonstrate the ability in three-dimensional tissue reconstruction.

These datasets has been utilized in previous studies (PASTE [1], BayesSpace [2], STAGATE [3] etc.), which demonstrate that these datasets are believed to be with proper sample size.

[1] Zeira, R., Land, M., Strzalkowski, A. & Raphael, B. Alignment and integration of spatial transcriptomics data. *Nature Methods* 19, 567-575 (2022).

[2] Zhao, E. et al. Spatial transcriptomics at subspot resolution with BayesSpace. *Nature Biotechnology* 39, 1375-1384 (2021).

[3] Dong, K. & Zhang, S. Deciphering spatial domains from spatially resolved transcriptomics with an adaptive graph attention auto-encoder. *Nature communications* 13, 1-12 (2022).

Data exclusions

We performed quality control of spatial resolved transcriptomics data based on the common used and pre-established criteria in this field.

Replication

Since the outcome from running the deep learning model may slightly differ across runs when the random seed is not fixed, we run the model ten times to evaluate its robustness.

Randomization

Randomization is not relevant to this study since we do not collect any new experimental data.

Blinding

Since all data used here are published in previous research, it is not possible to blind the investigations in reanalyzing the data. The allocation information are blinded to the computational algorithms during performance evaluation.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging